

Guide: The Correction-Coverage Gap Is Closed

Collatz proof project

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What Lean now proves

Suppose a positive natural Collatz orbit is unbounded. Then the sum of the reciprocals of all its successive values converges. This is a theorem about every hypothetical unbounded orbit, not about almost every starting seed.

The existing reciprocal theorem then gives bounded ideal correction. The previously unresolved case of an unbounded orbit with divergent reciprocal sum is now ruled out.

The key idea

An unbounded orbit never repeats a value. Two of its distinct values also cannot merge after the same number of additional steps: that would create a repeated value later.

Most small starting values map into a much smaller range after a suitable number of steps. Their odd-step counts have an exact binomial distribution, and high odd-step counts are rare. An orbit on which this map is injective cannot contain too many of these small starting values, because there are too few distinct endpoints to hold them.

Lean proves an explicit bound: among values below 32^m , any finite set from an unbounded orbit has size S satisfying

$$8^m S \leq 2 \cdot 216^m + 243^m.$$

Dividing the positive integers into shells from 32^m to 32^{m+1} turns this counting bound into a sum of two convergent geometric series.

What this adds to the proof program

The exact drift identity now gives

$$\frac{2^t}{3^{j_t}} \rightarrow 0,$$

where j_t counts odd steps. This means the additive logarithmic drift escapes to positive infinity. It does not imply a fixed positive gap in average odd-step density.

The bounded-correction machinery can now be applied to every hypothetical unbounded positive orbit. The next task is to find a contradiction using that machinery and the exact arithmetic constraints. A uniform absolute bound on all shifted corrections has not been proved.

Why this is not Collatz yet

Bounded correction is a restriction on an unbounded orbit; it is not an exclusion of one. Nontrivial cycles also remain a separate unresolved case. Finite cycles repeat values, so their infinite reciprocal series diverges; they do not satisfy the new summability hypothesis.

The mathematical implication is related to a quantitative 1999 result of Garcia and Tal. We make no novelty claim. The main project's proof uses its own existing bounds and adds no external mathematical axiom.

Verification

The Lean build and 113-declaration axiom audit pass, as do three exact finite controls. Run `lake build Collatz.OrbitSummability` from `lean/`. The technical paper and editable LaTeX sources provide the full argument, provenance, and precise remaining gap.